

Capital.com Senior Angular/Ionic & System Architect Interview Questions

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Answers

1. Angular Architecture & Internals

1.1. How does Angular's Dependency Injection (DI) system work under the hood, and how would you optimize the injector hierarchy in a large-scale application?

Under the Hood:

Angular's DI system is hierarchical and based on a tree of injectors that mirrors the component tree.

- **ElementInjector:** Created for each DOM element with a directive/component. It prioritizes local providers (`providers` array in Component/Directive decorator).
- **EnvironmentInjector (formerly ModuleInjector):** Configured via `NgModule` or `standalone` routes. It acts as the fallback if the ElementInjector doesn't find the token.
- **Resolution:** When a dependency is requested, Angular searches the current `ElementInjector`. If not found, it bubbles up to the parent `ElementInjector`. If it reaches the root Component without success, it switches to the `EnvironmentInjector` hierarchy, eventually reaching the `NullInjector` (which throws an error).

Optimization in Large-Scale Apps:

- **Tree-Shakable Providers:** Always prefer `{ providedIn: 'root' }` for services. This creates a singleton at the root level and allows unused services to be removed from the final bundle.
- **Avoid Component Providers:** Avoid adding services to a component's `providers` array unless you specifically need a separate instance for that component subtree. This prevents memory leaks and unintended state isolation.
- **Injection Tokens:** Use `InjectionToken` for configuration and primitive values to avoid collisions.
- **`inject()` function:** Use the modern `inject()` function for clearer, type-safe injection without constructor boilerplate.

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1.2. Explain the evolution of Angular's rendering engine from View Engine to Ivy. What specific architectural benefits does Ivy bring to a high-performance trading application?

Evolution:

- **View Engine (Legacy):** Compiled templates into separate `ngfactory.js` files. The compilation was slower, and the runtime required a larger framework core to interpret these factories.
- **Ivy (Current):** The "Locality" principle is key. Ivy compiles templates into instructions directly inside the component class (`ɵcmp`). This removes the need for separate factory files and allows for better static analysis.

Benefits for Trading Apps:

- **Bundle Size:** Because Ivy instructions are tree-shakable, the framework itself becomes smaller. Unused Angular features are stripped out, resulting in faster initial loads (TTFB/FCP).
- **Memory Efficiency:** Ivy generates DOM nodes more efficiently and allocates less memory for component metadata. In a trading dashboard with hundreds of blinking tickers, this reduces garbage collection pressure, leading to smoother 60fps animations.
- **Debugging:** Ivy enables standard browser console debugging (e.g., accessing component state via `ng.getComponent($0)`), crucial for diagnosing real-time data flow issues.

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1.3. Compare and contrast `ChangeDetectionStrategy.Default` and `ChangeDetectionStrategy.OnPush`. How do Angular Signals change this paradigm in recent versions?

Comparison:

- **Default:** Angular checks every component in the tree, from top to bottom, whenever *any* async event occurs (click, timer, XHR). This guarantees UI consistency but is computationally expensive for frequent updates.
- **OnPush:** Angular only checks the component if:
 1. An `@Input()` reference changes (requires immutability).
 2. An event originates from the component or its children.
 3. The `AsyncPipe` emits a new value.

4. `ChangeDetectorRef.markForCheck()` is manually called.

Impact of Signals:

Signals shift the paradigm from **"Check everything to see what changed"** to **"Tell exactly what changed"**.

- **Fine-Grained Reactivity:** Signals establish a direct dependency graph. When a signal updates, only the specific DOM nodes (text, classes) that depend on it are updated.
- **Zoneless:** Signals pave the way for removing `zone.js`. This eliminates the overhead of monkey-patching global browser APIs, significantly improving performance in data-heavy apps where `zone.js` often triggers unnecessary change detection cycles.

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1.3.1. How would you debug a scenario where the UI is not updating despite the data changing in an `OnPush` component?

Debugging Steps:

1. **Check Object Mutability:** The most common cause is mutating an object (e.g., `user.name = 'New'`) instead of creating a new reference. `OnPush` relies on reference equality (`===`).
2. **Verify Trigger Source:** Did the update happen outside Angular's zone (e.g., a third-party WebSocket library callback)? If so, Angular doesn't know it happened. Wrap the update in `NgZone.run()` or manually call `ChangeDetectorRef.markForCheck()`.

3. **Angular DevTools:** Use the profiler to see if the change detection cycle actually ran for that component.
4. **Console Log in Template:** A temporary hack is to call a function in the template `{{ logRender() }}` to see if the view is being re-evaluated.

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1.4. In a large-scale enterprise application, how would you structure modules and lazy loading to ensure the fastest possible Time to First Byte (TTFB) and First Contentful Paint (FCP)?

Structure & Optimization:

- **Feature-Based Architecture:** Group code by domain (e.g., `TradeModule`, `PortfolioModule`) rather than type (e.g., `Components`, `Services`).
- **Lazy Loading:** strictly lazy load all non-critical routes using `loadChildren` or standalone component lazy routes.
- **Deferrable Views (`@defer`):** Use the new `@defer` block to delay loading heavy components (like complex trading charts) until they are in the viewport or interacted with.
- **Preloading Strategy:** Implement a custom `PreloadingStrategy` to load critical feature chunks in the background *immediately after* the initial render, ensuring instant navigation later without blocking the initial paint.
- **Standalone Components:** Migrating to standalone components reduces the overhead of `NgModule` compilation and enables more granular code-splitting.

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1.5. Describe the concept of Micro-frontends in Angular using Module Federation. What are the trade-offs when implementing this in a financial platform?

Concept:

Micro-frontends allow splitting a monolithic app into smaller, independently deployable applications. Webpack Module Federation allows these apps to share a single runtime (e.g., a single instance of Angular Core and RxJS) to avoid payload duplication, while still loading code dynamically at runtime.

Trade-offs in FinTech:

- **Pros:**

- **Independent Deployment:** Different teams (e.g., "Payments", "Trading", "KYC") can deploy at their own pace without coordinating a massive release.
- **Fault Isolation:** A bug in the "News Feed" widget is less likely to crash the critical "Trading Ticket".

- **Cons:**

- **Complexity:** Orchestrating version compatibility (e.g., ensuring all remotes use compatible Angular versions) is difficult.
- **Performance:** Initial load can suffer due to the overhead of fetching remote entry files and negotiating shared dependencies.
- **State Management:** Sharing global state (like a WebSocket connection for prices) between isolated micro-frontends requires a well-designed,

neutral communication layer (e.g., a shared library or window-based bus), which adds architectural complexity.

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2. Ionic & Hybrid Mobile Development

2.1. Explain the bridge mechanism in Capacitor vs. Cordova. How does communication between the JavaScript layer and the Native layer impact performance?

The Bridge Mechanism:

- **Cordova:** Relies on a web view bridge that serializes messages between JavaScript and Native code. Historically, this involved URL-scheme hacking or `iframe` bridges, which introduced significant latency. Every plugin call is an asynchronous message passing event.
- **Capacitor:** Modernizes this by injecting a native JavaScript interface directly into the WebView (`window.Capacitor`). It uses a more efficient message-passing channel that allows for batched execution and faster data serialization. Capacitor also runs more "native" code by default, keeping the JS layer thinner.

Performance Impact:

The communication bridge is a bottleneck. Frequent crossing of this bridge (e.g., streaming accelerometer data at 60Hz) causes serialization/deserialization overhead on the main thread, leading to UI jank.

- **Mitigation:** Batch native calls where possible, or write a custom native plugin that handles high-frequency data (like price ticks) entirely on the

native side and only sends summarized updates to the WebView.

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2.2. How do you handle complex navigation state in an Ionic application that mirrors a web platform?

Handling Navigation:

Ionic uses `IonRouterOutlet` which integrates with the Angular Router but adds a stack-based navigation history (Push/Pop) typical of native apps, unlike the browser's linear history.

- **Synchronization:** The challenge is keeping the URL in sync with the native stack. Use absolute paths for top-level tabs and relative paths for drill-down details.
- **State Preservation:** Use `RouteReuseStrategy` to detach and store the state of complex views (like a trading dashboard) when navigating forward, and reattach it when navigating back, preserving scroll position and filled forms.
- **Split Pane:** For tablet/desktop, use `ion-split-pane` to map side-menus to auxiliary routes, ensuring deep-linking works correctly (e.g., `/market/crypto/BTC` opens the market tab, crypto category, and BTC details).

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2.3. What strategies would you employ to optimize the startup time of an Ionic application on older Android devices?

Optimization Strategies:

1. **Reduce Bundle Size:** Aggressive tree-shaking and lazy loading of modules. Ensure the initial chunk only contains the login/splash screen and core vendors.
2. **Server-Side Rendering (SSR) / Static Site Generation (SSG):** While harder in a hybrid app, pre-rendering the app shell (App Shell Pattern) ensures the user sees the UI structure immediately while JS downloads.
3. **Native Splash Screen:** Use the Capacitor Splash Screen API to hide the WebView initialization flickering. Configure `launchShowDuration: 0` and manually call `SplashScreen.hide()` only after the Angular app has fully rendered the first view (using `AfterViewInit`).
4. **Web View Warm-up:** Ensure the WebView is instantiated as early as possible in the native activity lifecycle.

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2.4. Discuss the differences between running an app as a PWA vs. a native store app. How do you handle offline capabilities and data synchronization in a trading context?

PWA vs. Native:

- **PWA:** Distribution via URL, instant updates (no app store review), limited access to native APIs (no FaceID, limited background processing), storage quotas are stricter.
- **Native (Capacitor):** Full access to Native SDKs, better performance (warm webview), persistent storage, push notifications reliability.

Offline & Sync:

- **Storage:** Use `SQLite` (via Capacitor community plugin) instead of LocalStorage/IndexedDB for reliable, persistent, relational data storage on the device.
- **Queueing Pattern:** When offline, trading actions (buy/sell) are validated locally (optimistic UI) and added to a persisted Redux/NgRx "Outbox" queue.
- **Sync:** Monitor `Network.addListener()`. When connection is restored, replay the queue sequentially.

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2.4.1. How would you resolve data conflicts when a user performs trades offline and then reconnects?

Conflict Resolution:

In trading, "Last Write Wins" is dangerous.

1. **Optimistic Updates:** Show the trade as "Pending" in the UI immediately.
2. **Version Vectors / Timestamps:** Send the trade with the client-side timestamp.
3. **Server Validation:** The server rejects the trade if the price has moved significantly beyond the user's "slippage tolerance" during the offline period.
4. **User Feedback:** If rejected, notify the user: "Market moved while you were offline. Trade for BTC at \$50k failed. Current price: \$52k. Retry?" Never automatically execute an old trade at a new price.

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2.5. How do you manage hardware-specific features (biometrics, secure storage) securely in an Ionic app?

Security Implementation:

- **Biometrics:** Use `capacitor-community/native-biometric`. Never store passwords directly. Instead, store a refresh token or a session encryption key in the device's **Secure Enclave** (Keychain on iOS, Keystore on Android). The biometric prompt simply unlocks this secure storage.
- **Secure Storage:** Do not use `LocalStorage` (it's plain text). Use `ionic-secure-storage` or `capacitor-secure-storage-plugin` which wraps the native OS secure storage mechanisms.
- **Memory Hygiene:** Clear sensitive variables from memory (JS garbage collection is lazy) when the app goes to the background (`App.addListener('appStateChange')`) to prevent data leakage in task switcher snapshots.

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3. RxJS & Reactive Programming

3.1. Real-time data is critical for Capital.com. How would you manage multiple WebSocket streams for different financial instruments using RxJS to ensure memory safety and performance?

Management Strategy:

- **Multiplexing:** Instead of opening a unique WebSocket connection for every financial instrument (which kills the browser limit), use a single socket connection and "multiplex" messages.
- **RxJS Implementation:** Use `websocket` from `rxjs/webSocket`. Create a central `socket$` subject.
- **Operator `multiplex()`:** Use the `multiplex` operator to filter messages for specific tickers. It handles sending the "subscribe" message when the first observer subscribes and the "unsubscribe" message when the last one unsubscribes.
- **Memory Safety:** Ensure strict unsubscription logic. In Angular components, use the `AsyncPipe` (which unsubscribes automatically) or `takeUntil(destroy$)`.
- **Performance:** Use `shareReplay({ bufferSize: 1, refCount: true })`. This multicasts the latest price to all subscribers (e.g., chart, order book, header ticker) without creating redundant socket subscriptions/connections.

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3.2. Explain the difference between `switchMap`, `mergeMap`, `concatMap`, and `exhaustMap`. Give a concrete example of when to use each in a trading application.

Operator Comparison:

- **`switchMap` (The "Latest" Strategy):** Cancels the previous inner observable if a new value arrives.

- *Example: **Typeahead Search***. If the user types "APP" and then "APPLE", you want to cancel the request for "APP" and only show results for "APPLE".
- **mergeMap** (The "Parallel" Strategy): Subscribes to every new inner observable immediately and handles them concurrently.
 - *Example: **Price Tickers***. If you receive price updates for AAPL, TSLA, and BTC simultaneously, you want to process and display all of them, not cancel one for the other.
- **concatMap** (The "Queue" Strategy): Waits for the previous inner observable to complete before starting the next.
 - *Example: **Order Execution***. You must process "Approve Wallet" before "Execute Trade". Order matters strictly.
- **exhaustMap** (The "Ignore" Strategy): Ignores new emissions while the current inner observable is active.
 - *Example: **"Buy" Button***. If a user frantically clicks "Buy", you execute the first click and ignore the rest until the trade confirms, preventing accidental double-spending.

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3.3. How do you handle error isolation in nested observable streams? If one stock ticker stream fails, how do you ensure the entire dashboard doesn't crash?

Error Isolation:

In RxJS, if an error reaches the main subscriber, the entire stream completes (dies). To prevent a single failing ticker from killing the whole dashboard:

- **Strategy:** Catch errors **inside** the inner observable (flattening operator).
- **Implementation:**

```
const dashboard$ = tickerIds$.pipe(
  mergeMap(id => getPriceStream(id).pipe(
    // Isolate error here!
    catchError(err => {
      console.error(`Ticker ${id} failed`, err);
      return of({ id, status: 'unavailable' }); //
    })
  ))
);
```

- **Result:** If `getPriceStream('BTC')` fails, only that specific inner stream is replaced by the fallback. The main `dashboard$` stream continues to process `AAPL`, `GOOG`, etc.

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3.4. What are "Hot" and "Cold" Observables? Why is this distinction crucial when implementing a multicasted data store?

Distinction:

- **Cold Observable:** The producer is created *inside* the observable for *each* subscriber.
 - *Example:* `HttpClient.get()`. If 3 components subscribe to `get('/prices')`, the browser sends 3 separate network requests.

- **Hot Observable:** The producer is created *outside* and shared.
 - *Example:* `new Subject()`, DOM events, WebSocket. If 3 components subscribe, they all listen to the *same* running stream.

Crucial for Data Store:

A state management store (like NgRx or a Service-with-Subject) must be **Hot**.

- If your store selects state as a Cold Observable, every component (Header, Sidebar, Dashboard) that subscribes to `selectUser()` would trigger the underlying calculation or API call again.
- You must use `shareReplay` or `BehaviorSubject` to ensure the state is calculated once and multicasted to all consumers to avoid performance degradation and inconsistent state.

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3.5. Describe how you would implement a retry strategy with exponential backoff for failed network requests using RxJS operators.

Strategy:

Instead of a simple `retry(3)` which retries immediately, use `retryWhen` (or `retry` with config in RxJS 7+) to delay the attempts.

Implementation (RxJS 7+):

```
data$.pipe(  
  retry({  
    count: 3,  
    delay: (error, retryCount) => {
```

```

        // Exponential backoff: 1s, 2s, 4s...
        return timer(Math.pow(2, retryCount) * 1000);
    }
}))
);

```

Implementation (Legacy `retryWhen`):

```

data$.pipe(
  retryWhen(errors =>
    errors.pipe(
      // Merge map to use the index (i) for exponential
      calculation
      mergeMap((error, i) => {
        const retryAttempt = i + 1;
        if (retryAttempt > 3) {
          throw error; // Give up after 3 tries
        }
        return timer(Math.pow(2, retryAttempt) * 1000);
      })
    )
  )
);

```

This prevents hammering the server during an outage (Thundering Herd problem).

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4. State Management & Data Flow

4.1. Compare NgRx, Akita, and Elf. Which would you choose for a high-frequency trading dashboard and why?

Comparison:

- **NgRx:** The industry standard. Uses Redux pattern (Actions, Reducers, Effects, Selectors). Extremely robust, boilerplate-heavy, best for complex enterprise apps where traceability is key.
- **Akita:** OOP-based state management. Easier for developers coming from backend backgrounds. Less boilerplate, supports entities and queries out of the box. Deprecated in favor of Elf.
- **Elf:** The successor to Akita. Built on top of RxJS, tree-shakable, modular. Very lightweight.

Choice for High-Frequency Trading (HFT):

I would choose **Elf** or a custom **Service-with-Subject** architecture (using `BehaviorSubject` and `scan` operator).

- **Reason:** NgRx adds a significant overhead (action dispatch -> reducer -> selector) for every single price tick. In HFT with 100 updates/sec, this CPU overhead adds up. Elf or raw RxJS services offer a "thinner" abstraction layer, allowing for faster updates and less garbage collection, which is critical for maintaining 60fps in a data-intensive grid.

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4.2. How do you prevent "state bloat" in a long-running Single Page Application?

Prevention Strategies:

1. **Local State vs. Global State:** Don't put everything in the store. Form state, open/close UI toggles, and temporary filters should remain in the component or a local `ComponentStore`. Only shared data belongs in the global store.
2. **Entity Management:** Use `@ngrx/entity` to normalize data. Store objects by ID in a dictionary `{ [id]: entity }` rather than an array. This prevents duplication and makes updates $O(1)$.
3. **Cache Invalidation:** Implement a meta-reducer or effect that clears specific state slices on logout or module destruction.
4. **Pagination/Infinite Scroll:** Never store "all history". Store only the currently viewed window of data.

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4.3. Explain the concept of Selectors in NgRx. How can memoization optimization impact the performance of a grid rendering thousands of live prices?

Selectors & Memoization:

Selectors are pure functions that slice a piece of state from the store.

- **Memoization:** NgRx selectors cache the last result. If the input arguments (the state slice) haven't changed (reference check), the selector returns the cached calculation immediately without re-executing the transformation logic.

Impact on Trading Grid:

Imagine a grid calculating `Profit/Loss = (CurrentPrice - OpenPrice) * Volume`.

- Without memoization: Every time *any* part of the state changes (e.g., a unrelated notification arrives), the P/L calculation would re-run for all 1000 rows.
- With memoization: The calculation only re-runs if `CurrentPrice` or `OpenPrice` specifically changes for that row. This prevents thousands of unnecessary mathematical operations per second, keeping the UI responsive.

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4.4. How would you architect the state management to handle multi-tab synchronization (e.g., user changes theme or language in one tab, and it updates in others)?

Architecture:

1. **Storage Event:** The `localStorage` API fires a `storage` event in other tabs whenever a key is modified.

2. Implementation:

- Create a `CrossTabSyncService`.
- Listen to `window.addEventListener('storage', (event) => { ... })`.
- When the "Theme" action is dispatched in Tab A, save it to LocalStorage.
- Tab B receives the storage event.
- The service in Tab B reads the new value and dispatches a `[Theme]` `Update from Storage` action to its own internal store.

3. **Broadcast Channel API:** For more complex data (not just strings), use the `BroadcastChannel` API, which allows sending structured JSON objects between browsing contexts (tabs/windows/iframes) on the same origin.

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5. System Design & Performance (FinTech Context)

5.1. Design a high-level architecture for a real-time trading frontend that consumes millions of price updates per second. How do you handle throttling and debouncing without losing critical last-price accuracy?

Architecture:

1. **Transport Layer:** WebSockets (using `protobuf` for binary data serialization to reduce payload size).
2. **Worker Layer:** A dedicated **Web Worker** parses the incoming binary stream. This keeps the main thread free for rendering/user interaction.
3. **State Layer:** The Worker posts processed updates to a shared `SharedArrayBuffer` or sends batched updates (e.g., every 16ms/60fps) to the main thread.

Throttling vs. Debouncing:

- **Never Debounce Prices:** Debouncing waits for a pause. In a volatile market, prices change constantly, so a debouncer might never emit!
- **Throttling (AuditTime):** Use `auditTime(16)` or `animationFrameScheduler`. This ignores intermediate values but

guarantees that the *very last* emitted value in that time window is the one processed.

- **Visual Throttling:** The internal data model can update 1000 times/sec, but the UI grid only re-renders when `requestAnimationFrame` fires, ensuring the user always sees the latest price on the next paint without freezing the browser.

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5.2. How do you secure a frontend application against XSS and CSRF attacks, specifically when dealing with financial transactions and JWT tokens?

Security Measures:

- **XSS (Cross-Site Scripting):**
 - **Angular's Default Sanitization:** Angular automatically sanitizes values in property binding. Never use `bypassSecurityTrustHtml` unless absolutely necessary and audited.
 - **Content Security Policy (CSP):** Implement a strict CSP header (`script-src 'self'`) to prevent execution of unauthorized scripts injected via malicious ads or compromised CDNs.
- **CSRF (Cross-Site Request Forgery):**
 - **SameSite Cookies:** Set the `SameSite=Strict` attribute on authentication cookies.
 - **Anti-CSRF Tokens:** The backend injects a unique token in the cookie; Angular's `HttpClient` automatically reads this token (e.g., `XSRF-`

`TOKEN`) and sends it back in a custom header (`X-XSRF-TOKEN`) for mutation requests.

- **JWT Storage:**

- **Ideally:** Do NOT store JWTs in `localStorage` (accessible by XSS). Store them in **HttpOnly, Secure Cookies**.
- **If LocalStorage is mandatory:** Use short-lived tokens (5 min) and implement a sliding session with a refresh token rotation mechanism.

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5.3. Explain how you would implement a robust logging and monitoring system for client-side errors that respects user privacy and GDPR regulations.

Implementation:

1. **Global Error Handler:** Implement Angular's `ErrorHandler` class to catch all unhandled exceptions.
2. **PII Scrubbing:** Before sending the error report:
 - Redact sensitive patterns (Credit Card Regex, Email Regex).
 - Do not send the entire `store` state if it contains user balances or names.
 - Use anonymized User IDs, not real names.
3. **Sampling:** For high-traffic apps, don't log every error. Use a sampling rate (e.g., log 10% of errors) to prevent flooding the logs, unless it's a critical "Blocker" severity.

4. **Tools:** Use Sentry or Datadog with "Data Scrubbing" rules configured on the server-side as a fail-safe.

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5.4. What is your approach to handling "Race Conditions" in the frontend when multiple asynchronous processes depend on shared resources?

Approach:

1. **SwitchMap:** The "Silver Bullet" for most race conditions (e.g., autocomplete). It cancels the previous pending request when a new one starts.
2. **Locks / Mutexes:** For critical operations (like "Place Order"), implement a simple boolean lock:

```
if (this.isPlacingOrder) return;
this.isPlacingOrder = true;
this.tradeService.placeOrder().pipe(
  finalize(() => this.isPlacingOrder = false)
).subscribe();
```

3. **Versioning:** When displaying data, the server should return a version number. If the frontend tries to update a trade with `version: 5` but the server has `version: 6`, the server rejects it (Optimistic Concurrency Control), and the frontend must re-fetch and re-prompt the user.

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5.5. How would you design a feature-flagging system to roll out a new trading interface to 5% of high-net-worth users first?

Design:

1. **Service:** Create a `FeatureFlagService` that initializes *before* the app bootstraps (`APP_INITIALIZER`).
2. **Resolution:**
 - Fetch flags from an API endpoint: `GET /api/flags` .
 - The backend determines the user's segment ("High Net Worth") and returns: `{ "new-trading-ui": true }` .

3. **Directive:** Create a structural directive `*appFeature="'new-trading-ui'"`:

```
@Input() set appFeature(flag: string) {
  if (this.featureService.isEnabled(flag)) {

this.viewContainer.createEmbeddedView(this.templateRef);
  } else {
    this.viewContainer.clear();
  }
}
```

4. **Route Guards:** Use `CanMatch` (Angular 14+) guards to load entirely different lazy-loaded modules based on the flag, keeping the bundle size small for users who don't have the feature.

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6. Testing, CI/CD & Best Practices

6.1. Differentiate between Unit, Integration, and E2E testing in the context of an Angular/Ionic app. What is your preferred stack (e.g., Jest vs. Jasmine, Cypress vs. Playwright)?

Differentiation:

- **Unit Testing:** Tests individual components, services, or functions in isolation. All external dependencies are mocked.
 - *Example:* Testing a `PriceCalculatorService.calculateProfit()` method with mocked input data.
 - *Goal:* Verify business logic correctness. Fast (milliseconds), no DOM, no network.
- **Integration Testing:** Tests how multiple units work together. Some real dependencies, some mocked (e.g., real Angular DI but mocked HTTP).
 - *Example:* Testing a `TradingTicketComponent` that injects `OrderService` (real) and `HttpClient` (mocked via `HttpClientTestingModule`).
 - *Goal:* Verify component-service interaction. Medium speed.
- **E2E (End-to-End) Testing:** Tests the entire application from the user's perspective in a real browser. No mocks.
 - *Example:* Opening the app, logging in, navigating to "BTC/USD", placing a buy order, and verifying the order appears in the "Portfolio" tab.
 - *Goal:* Verify critical user flows work. Slow (seconds per test).

Preferred Stack:

- **Unit/Integration: Jest** over Jasmine. Jest is 3-5x faster due to parallel test execution and better caching. It has a superior mocking API (`jest.mock()`) and built-in code coverage.
- **E2E: Playwright** over Cypress.
 - *Reasons:*
 - True multi-browser support (Chromium, Firefox, WebKit) vs. Cypress's limited Firefox/Edge support.
 - Better network interception (service workers, WebSockets).
 - Native mobile browser testing (critical for Ionic).
 - Faster and more reliable auto-waiting mechanism.

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6.2. How do you verify the correctness of financial calculations on the frontend?

Verification Strategy:

1. **Property-Based Testing:** Use libraries like `fast-check` to generate thousands of random inputs and verify invariants (e.g., `profit + loss === 0` for closed positions).
2. **Test Against Known Values:** For critical formulas (leverage, margin requirements), hard-code test cases with values verified by the finance team or regulatory documentation.
3. **Backend Parity Tests:** Run the same calculation in both frontend (TypeScript) and backend (e.g., Python). Assert that

`frontend.calculate(X) === backend.calculate(X)` within a floating-point epsilon (`0.0001`).

4. **Precision Libraries:** Use `decimal.js` or `big.js` instead of JavaScript's native `Number` for currency calculations to avoid floating-point errors (e.g., `0.1 + 0.2 !== 0.3`).
5. **Snapshot Testing:** For complex UI outputs (like a P/L report table), use Jest snapshots. Any unintended change in the rendered calculation output will fail the test.

Example Test:

```
import Decimal from 'decimal.js';

describe('Profit Calculation', () => {
  it('should calculate profit with precision', () => {
    const openPrice = new Decimal('1.23456');
    const closePrice = new Decimal('1.23789');
    const volume = new Decimal('1000');

    const profit =
closePrice.minus(openPrice).times(volume);

    expect(profit.toFixed(2)).toBe('3.33'); //
Expected: $3.33
  });
});
```

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6.3. Describe a CI/CD pipeline for a hybrid Ionic app. How do you automate deployments to both the Apple App Store and Google Play Store?

CI/CD Pipeline Architecture:

Stage 1: Build & Test (GitHub Actions / GitLab CI)

1. **Trigger:** Push to `main` or `release/*` branch.
2. **Steps:**
 - Install dependencies (`npm ci`).
 - Lint (`eslint`, `stylelint`).
 - Unit tests (`npm run test:ci` with Jest, headless, code coverage > 80%).
 - Build web app (`ionic build --prod`).
 - Build native apps:
 - Android: `ionic capacitor build android --release`
 - iOS: `ionic capacitor build ios --release`

Stage 2: E2E Testing

- Run Playwright tests against the built web app (`npm run e2e`).
- For native: Use **Appium** or **Detox** to run critical smoke tests on emulators/simulators.

Stage 3: Code Signing & Deployment

Android (Google Play Store):

1. **Keystore Secret:** Store the signing keystore (`.jks` file) and passwords in CI environment secrets.
2. **Build Signed APK/AAB:**

```
./gradlew bundleRelease \
-
Pandroid.injected.signing.store.file=$KEYSTORE_FILE \
-
Pandroid.injected.signing.store.password=$KEYSTORE_PASSWORD \
-Pandroid.injected.signing.key.alias=$KEY_ALIAS \
-
Pandroid.injected.signing.key.password=$KEY_PASSWORD
```

3. **Upload to Play Store:** Use Fastlane (`supply` action):

```
lane :deploy_android do
  supply(
    track: 'internal', # internal → alpha → beta →
production
    aab:
'android/app/build/outputs/bundle/release/app-
release.aab'
  )
end
```

iOS (App Store):

1. **Certificates & Provisioning Profiles:** Use **Fastlane Match** to sync signing certificates from a private Git repo (encrypted).

2. Build & Archive:

```
fastlane gym --scheme "App" --export_method app-store
```

3. Upload to App Store Connect:

```
lane :deploy_ios do
  pilot( # TestFlight
    ipa: 'App.ipa',
    skip_waiting_for_build_processing: true
  )
end
```

4. **Auto-Submit for Review (Optional):** Use `deliver` to automatically submit the build for App Store review after TestFlight validation.

Stage 4: Notifications

- Send Slack/Teams notification with build status and version number.
- Tag the Git commit with the release version (`v1.2.3`).

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6.4. How do you enforce code quality and architectural standards in a team of 20+ frontend developers?

Enforcement Strategy:

1. Automated Code Quality Gates (Non-Negotiable)

- **PR-Required Checks:** GitHub/GitLab branch protection rules mandate that all automated checks pass before merging.

- **SonarQube:** Integrate with CI to enforce:
 - Code Coverage: Minimum 80% line coverage.
 - Code Smells: Maximum "Technical Debt" threshold (e.g., no "A" rated issues unresolved).
 - Security: Zero critical vulnerabilities (dependency scanning with `npm audit`, Snyk).

2. Architectural Decision Records (ADRs)

- Document major architectural decisions (e.g., "Why we chose NgRx over Akita") in Markdown files in the repo (`docs/adr/001-state-management.md`).
- Enforce a rule: Any deviation from established patterns requires a new ADR, reviewed in an architecture guild meeting.

3. Mono-repo with Nx (if applicable)

- Use **Nx** to enforce module boundaries (e.g., `trading` module cannot import from `kyc` module).
- CI fails if a developer violates the dependency graph.

4. Code Reviews with Checklists

- Use GitHub PR templates with a checklist:
 - ☐ No `any` types (enforce `strict: true` in `tsconfig.json`).
 - ☐ All observables unsubscribed (or use `AsyncPipe`).
 - ☐ Unit tests added for new logic.
 - ☐ No hardcoded API URLs (use environment files).

5. Pair Programming for Critical Features

- For high-risk changes (e.g., payment flows), mandate pair programming or "mob" code reviews with 2+ senior engineers.

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6.4.1. What automated tools would you include in the pre-commit hooks?

Pre-Commit Hook Tools (via Husky + lint-staged):

Setup:

```
// package.json
{
  "husky": {
    "hooks": {
      "pre-commit": "lint-staged"
    }
  },
  "lint-staged": {
    "*.ts,*.js": [
      "eslint --fix",
      "prettier --write"
    ],
    "*.scss,*.css": [
      "stylelint --fix"
    ],
    "*.ts,*.spec.ts": [
      "jest --bail --findRelatedTests"
    ]
  }
}
```

```

    }
  }
}

```

Tools:

1. ESLint (`eslint --fix`):

- Auto-fix formatting issues (trailing commas, semicolons).
- Enforce rules: `no-console`, `@typescript-eslint/no-explicit-any`, `rxjs/no-ignored-subscription`.

2. Prettier (`prettier --write`):

- Enforce consistent code formatting (line width, tabs vs. spaces).
- Non-negotiable: Code is auto-formatted, no debates about style.

3. Stylelint (`stylelint --fix`):

- Enforce SCSS/CSS best practices (e.g., no `!important`, BEM naming convention).

4. Jest (Related Tests Only):

- Run unit tests *only* for the files being committed (`--findRelatedTests`).
- Prevents committing broken code without slowing down every commit.

5. Commitlint:

- Enforce conventional commit messages: `feat:`, `fix:`, `chore:`.
- Enables automated changelog generation.

6. Type Checking (`tsc --noEmit`):

- Run TypeScript compiler in no-emit mode to catch type errors before commit.

Why Pre-Commit (not Pre-Push)?

Catching issues *before* the commit is pushed prevents polluting the shared branch history with "fix lint" commits. It's faster feedback for the developer.

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